TELONE CENTRE FOR LEARNING

TECHNICAL DEPARTMENT

DIPLOMA IN TELECOMMINICATIONS ENGINEERING

AN AUTOAMATED GAS DETECTION AND VENTILATION CONTROL SYSTEM FOR FUEL DEPOTS

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A research project submitted in partial fulfilment of the requirements for Diploma in Telecommunications Engineering, September 2026.

Declaration Page

I declare that this is my original work except where sources have been cited and acknowledged. The work has never been submitted, nor will it ever be submitted to another college or university for the award of a diploma.

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Student's Full Name   Student's Signature   (Date)

APPROVAL STATEMENT

This project titled "**____**" has been submitted in partial fulfillment of the requirements for the award of the Diploma in Telecommunications Engineering.

The undersigned certify that the work presented in this report was carried out by **____ (Student Name & Reg Number)** under our supervision and guidance. **_**

We hereby approve this project report for submission and examination.

Supervisor's Name:___________________________________

Signature:____

Date: ____

Head of Department: __________________________________

Signature: ____

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CHAPTER 1 : INTRODUCTION

1.1 Background

Fuel storage depots form a critical component of Zimbabwe's petroleum supply chain, facilitating the storage and distribution of petroleum products such as petrol, diesel, and liquefied petroleum gas (LPG) to industries, businesses, and households. Despite their importance, these facilities pose significant safety risks due to the potential accumulation of flammable hydrocarbon vapours.

Leakages from storage tanks, pipelines or valves can result in the rapid buildup of hazardous gases, which may lead to fires, explosions and serious health risks for personnel. Globally, automated gas detection and ventilation systems have been widely implemented to mitigate such risks. These systems continuously monitor gas concentrations, trigger alarms, and activate ventilation mechanisms to disperse hazardous vapours before they reach critical levels.

In Zimbabwe, however many fuel depots still rely on manual inspection methods or basic gas detection devices that provide limited coverage and delayed response. Additionally, continuously operating ventilation systems are often used leading to excessive energy consumption—an unsustainable approach given the country's frequent power constraints.

This project focuses on the development and simulation of an automated gas detection and ventilation control system through a prototype model. The prototype demonstrates how gas sensors, control logic, alarm mechanism and fans can be integrated to enhance safety and energy efficiency. Although not intended for immediate industrial deployment, the system serves as a proof-of-concept, illustrating the practical application of automation and telecommunications engineering principles in improving fuel depot safety.

1.2 Problem Statement

Fuel storage depots in Zimbabwe lack cost-effective and automated systems for detecting and mitigating gas leaks. Existing approaches rely heavily on manual inspections or basic alarm systems that are not integrated with ventilation control or intelligent response mechanisms.

This limitation increases the risk of fire and explosion, compromises worker safety, and leads to inefficient energy usage due to continuous ventilation operation. Therefore, there is a need to develop a prototype system that demonstrates how gas detection, automated ventilation and alarm systems can be integrated into a simple, reliable and efficient solution.

1.3 Aim

The aim of this project is to design and simulate an automated gas detection and ventilation control system using a prototype model.

The system will:

- Detect hydrocarbon vapours in real time
- Automatically activate ventilation systems when gas levels exceed predefined safety thresholds
- Trigger local alarm mechanisms (buzzer and LED indicators) to alert personnel
- Validate system functionality through simulation results

1.4 Objectives

The specific objectives of the project are:

- To reduce the risk of fire and explosion through early detection of hydrocarbon gases and automated mitigation.
- To enhance operational safety in fuel storage facilities using continuous gas monitoring systems.
- To minimize human intervention and error by implementing automated detection and control mechanisms.
- To improve system response time through real-time sensing and automatic ventilation activation.
- To develop a real-time alert system incorporating alarms and indicators for immediate hazard notification.

1.5 Research Questions

- How can hazardous gases in fuel storage depots be detected accurately and in real-time?
- Which types of gas sensors are most effective for detecting fuel-related gases such as LPG, methane, and petrol vapours?
- How can an automated system integrate gas detection with ventilation control to reduce gas concentration levels?

- What threshold levels should be set to trigger ventilation systems and alarms effectively?
- How can the response time of the system be optimized to prevent fire or explosion hazards?
- What role does ventilation play in mitigating gas accumulation in enclosed fuel storage environments?
- How can environmental factors (temperature, humidity, airflow) affect gas detection and ventilation efficiency?
- What control mechanisms can be implemented to automatically activate and deactivate ventilation systems?
- How reliable and cost-effective is an automated gas detection and ventilation system compared to manual monitoring methods?

1.6 Project Scope

- This project focuses on the design and implementation of an automated gas detection and ventilation control system for fuel storage depots.
- The system is intended to detect hazardous gases such as LPG, methane, and petrol vapours using appropriate gas sensors and continuously monitor their concentration levels in real-time.
- Upon detecting gas levels above a predefined threshold, the system will automatically activate a ventilation mechanism such as exhaust fans to reduce gas accumulation and improve safety.
- Additionally, the system will include alert features such as alarms or indicators to notify personnel of potential hazards.
- However, the project is limited to a prototype or small-scale implementation and does not extend to large-scale industrial deployment, integration with complex industrial control systems or compliance with all industrial regulatory standards.
- The final output will include a functional prototype, system documentation, and performance analysis.

1.7 Justification

- This project is justified by the need to improve safety through early detection of hazardous gases such as LPG and methane.
- The objectives of real-time monitoring, automated response and immediate alert systems directly address the problem of delayed and unreliable manual detection.
- By minimizing human intervention and improving response time the system reduces the risk of fire and explosions.
- Additionally, it provides a practical and cost-effective safety solution for small-scale environments while demonstrating the application of modern sensing and automation technologies.

1.8 Layout of Chapters

- **Chapter 1:** Introduction including background, problem statement, aim, objectives, research questions, scope and justification
- **Chapter 2:** Literature review on gas detection technologies, ventilation systems and alarm mechanisms
- **Chapter 3:** System design and simulation methodology, including block diagrams and control logic
- **Chapter 4:** Simulation results and performance evaluation

Chapter 5: Conclusions, challenges encountered and recommendations

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter looks at what other researchers have written about systems that automatically detect dangerous gases and control ventilation. It focuses on fuel storage areas. The review covers information from academic papers, industry reports and technical articles published between 2014 and 2026. This helps build the background knowledge needed for the new system proposed in this project. Because working with hazardous materials can be dangerous, there's a growing need for safety systems that work automatically. Khan and Abbasi (2016) found that most accidents at fuel-handling sites happen because flammable vapor builds up and isn't detected quickly enough and because there are no automatic systems to respond. This shows how important it is to have systems that can sense danger in real time and react on their own.

This need is especially strong in developing countries like Zimbabwe, where people still mostly rely on manual checks.

This review covers:

- Gas sensors (electrochemical and semiconductor types)
- How ventilation is controlled
- Alarm systems
- Automation using microcontrollers
- Monitoring systems connected to the Internet of Things (IoT)

2.2 Theoretical Background

2.2.1. How Flammable Gases Behave and When They Become Dangerous

Fuels like LPG (cooking gas), petrol vapor, and methane can catch fire or explode, but only if their concentration in the air is within a certain range. This range is defined by two limits:

· Lower Explosive Limit (LEL) – the smallest amount of gas in the air that can still catch fire. Below this, there's too little gas to burn.

· Upper Explosive Limit (UEL) – the highest amount of gas in the air that can still catch fire. Above this, there's too much gas and not enough oxygen to burn.

For LPG, the lower limit is about 1.8% gas in the air, and the upper limit is about 9.5% (Crowl and Louvar, 2019).

Safety systems are usually set to warn you when gas levels reach 10–25% of the lower limit. This gives you enough time to react before the gas becomes dangerous.

How Gases Spread in a Closed Space

Gases move through the air based on a physical rule called Fick's Law of Diffusion. How they spread also depends on:

- How well the space is ventilated
- The temperature
- Whether the gas is heavier or lighter than air

Heavier-than-air gases (like LPG) sink and collect near the floor. Lighter gases (like methane) rise toward the ceiling. This is important because it tells you where to place gas sensors for the system to work properly (Mannan, 2014).

2.2.2 How Gas Sensors Work

Gas sensors use different physical and chemical methods to detect dangerous gases.

i.)Semi-conductor sensors (like the MQ series)

These are cheap and common, making them good for building prototypes. They work by heating a metal oxide surface (usually tin dioxide). When gas touches this surface the sensor's electrical resistance changes. That change is turned into a voltage signal that rises or falls with the gas concentration. Examples include MQ-2, MQ-5 and MQ-135 (Mounir et al., 2020).

ii.) Infrared sensors (NDIR)

These work by shining infrared light through the air. Each type of gas absorbs light at a specific wavelength. These sensors are more accurate and reliable than semiconductor ones, but they are much more expensive. They are typically used in industrial settings (Lee et al., 2015).

iii.) Electrochemical sensors

These create a small electric current when a chemical reaction happens with the target gas. They are mainly used for detecting toxic gases like carbon monoxide.

Catalytic bead sensors (pellistors)

These measure the heat produced when a flammable gas burns on a catalytic surface. They are often used in areas where explosions are a serious risk (Williams, 2017).

Choosing the right sensor

You have to balance several factors: accuracy, how fast it responds, cost and how well it works in harsh environments. For a low-cost prototype that detects fuel vapours, semiconductor sensors are the best choice. Motla et al. (2021) showed that MQ-series sensors work well for monitoring fuel depots when connected to an Arduino-based system.

2.2.3Ventilation Principles In Hazardous Environments

In places where flammable or toxic gases might build up ventilation does two main jobs: it dilutes the gases (mixes them with fresh air so they become less concentrated) and removes them before they reach dangerous levels.

There are two main approaches:

i). Dilution ventilation – Brings in clean air to lower the overall gas concentration.

ii.) Local exhaust ventilation – Sucks up the gas right at the source, before it can spread.

How to calculate the ventilation you need

You can figure out the required airflow with this basic idea:

- The more gas being released, the more airflow you need.
- The lower you want the gas concentration to stay, the more airflow you need.

The actual formula is:

Airflow needed = (Gas release rate) ÷ (Safe gas limit – Gas already in incoming air)

(ASHRAE, 2019)

Why automatic ventilation is better

Systems that turn on ventilation only when a sensor detects gas are much better than systems that run all the time. A study by Huang et al. (2018) found that in enclosed industrial spaces automatic, sensor-triggered ventilation used up to 47% less energy than constantly running fans. And it was just as safe.

This is exactly why this project uses an energy-efficient design

2.2.4Microcontroller -Based Control Systems

Microcontrollers are small, cheap computer chips that control devices automatically. Two popular ones are the Arduino Uno (which uses the ATmega328P chip) and the PIC16F877A. They are widely used because they:

- Cost very little
- Are easy to program

Have built-in features like:

- ADC (analogue-to-digital converter) – reads changing voltage signals from sensors
- PWM (pulse-width modulation) – controls things like fan speed
- Digital I/O pins – turns things on or off (like lights, buzzers, or relays)

The Arduino platform is especially common in student projects and safety system prototypes. Banzi and Shiloh (2014) pointed out that it's great for quickly building and testing new ideas.

What the microcontroller does step by step:

i. It receives a changing voltage signal from the gas sensor (through its ADC).

ii. It compares that signal to a danger threshold value stored in its memory.

iii. Based on that comparison, it activates outputs like:

- Relays (switches to turn on fans or other high-power devices)
- Fans (to ventilate the area)
- Alarms (lights or buzzers to warn people)

So in simple terms: the microcontroller reads the sensor, checks if gas levels are dangerous and then automatically turns on fans and alarms if needed.

2.3 Review Of Existing Systems / Technologies

2.3.1 Traditional Manual Monitoring Methods

In the past fuel storage facilities checked for gas leaks by having trained workers walk around with handheld gas detectors at set times. This manual method costs less upfront because you don't need to buy expensive automated systems. However, it has serious problems:

- Checks happen too infrequently – many leaks can be missed between inspections.
- Human error – workers might make mistakes or miss a spot.
- Limited coverage – a person can't be everywhere at once.
- Slow response – by the time a leak is found, it may already be dangerous.
- A real-world example from Zimbabwe

Chikoto and Ndebele (2020) studied fuel depots in Zimbabwe and found that in many facilities, workers checked for gas leaks only every two to four hours. That might sound reasonable but here's the problem:

If a small leak starts right after a check, the gas concentration can reach explosive levels long before the next inspection happens. Two to four hours is far too slow to catch a dangerous buildup.

This is why automated, continuous monitoring is so important

2.3.2 Fixed Point Automated Gas Detection System

Fixed-point gas detection systems are a big step up from having people walk around with handheld detectors. Here's how they work:

- Sensors are mounted in fixed strategic spots around the facility (e.g., near the floor where heavy gases like LPG collect
- They run continuously sending data to a central control panel or alarm system.

Real-world examples of these systems include the Honeywell Sense point XCD and the Drager Polytron series (Honeywell, 2018; Dragerwerk AG, 2019).

These professional systems offer:

- 24/7 monitoring – no gaps between checks.
- Automatic alarms – they sound the moment gas is detected.
- Integration with ventilation – some can automatically turn on fans using relay outputs.

The Problem: They're Too Expensive for Many Places

These commercial systems cost a lot of money to buy and maintain – typically between $2,000 and $15,000 per sensor. That's per single detection point, not for the whole system.

For small and medium-sized fuel storage operators in sub-Saharan Africa, that's simply unaffordable (Ndlovu, 2022).

Even if they could afford the upfront cost, there are other issues:

- They don't work well with locally available parts or repairs.
- They need specially trained technicians to maintain them.

- This creates a sustainability problem in developing countries where such experts may not be available nearby.

This is exactly why a low-cost locally maintainable system is needed.

2.3.3 SCADA -Based Integrated Safety System

SCADA stands for Supervisory Control and Data Acquisition. Think of it as the most advanced all-in-one safety and control system for industrial settings. It combines several parts into one unified system:

- Sensor networks – to detect gases
- PLCs (Programmable Logic Controllers) – rugged computers that control machinery
- HMIs (Human-Machine Interfaces) – screens that let operators see what's happening
- Remote monitoring – ability to check conditions from a distance

What SCADA can do

A study by Igure et al. (2016) highlights SCADA's powerful features:

- Manage complex facilities with multiple safety zones
- Log real-time data continuously
- Analyse trends (e.g. "gas levels have been rising slowly for an hour")
- Send predictive maintenance alerts (e.g."this sensor may fail soon, replace it")

Real-world examples in the petroleum industry

Large-scale refineries and fuel distribution terminals use SCADA systems such as Wonderware and Rockwell FactoryTalk. These offer excellent safety and operational control.

The Problem for Zimbabwe

While SCADA systems work great for big, well-funded operations, they face major barriers in Zimbabwe's petroleum sector (Chinyanganya and Maposa, 2021):

- Huge financial cost – these systems are very expensive
- Need for high-speed internet or dedicated communication networks – these are not always available
- Need for expert system integrators – highly trained specialists are required to set them up

- Resources are not readily available in Zimbabwe

So, SCADA is too big, too expensive and too complex for the scale and resources available locally. This reinforces the need for a simpler more affordable solutions

2.3.4 IoT -Enabled and Wireless Gas Monitoring Systems

Wireless and Internet-Connected Gas Monitors (IoT)

The rise of the Internet of Things (IoT) — everyday devices connected to the internet — has made gas monitoring much more affordable. These systems use:

- Cheap, small microcontrollers (like Arduino)
- Wireless communication (Wi-Fi)
- Cloud computing (data stored online)

This combination provides remote monitoring and alerting that rivals expensive industrial SCADA systems but at a fraction of the cost (Atzori et al., 2017).

Real examples from Africa

- Ghana: Osei and Asante (2023) built a ZigBee-based gas monitoring network at a fuel depot. It detected gas in under 5 seconds and sent real-time SMS alerts and a web dashboard update.
- Kenya: Munyoro et al. (2022) built a system using Arduino and an ESP8266 (a cheap Wi-Fi chip) for a fuel storage application. The cost was under $150 per monitoring node, and it was over 92% as accurate as a much more expensive reference sensor.

These studies prove that low-cost IoT approaches work well in sub-Saharan African countries like Zimbabwe.

2.3.5Ventilation Control Integration

Since 2014, researchers have made big progress in linking gas detectors to automatic fans.

Smarter than just on/off

· Fuzzy logic system (Li et al., 2018): Designed for underground car parks. Instead of just turning fans fully on or off, it adjusts fan speed gradually based on how much gas is present. This saves energy while keeping the area safe.

· PID control system (Zhao et al., 2020): Designed for chemical storage warehouses. It uses closed-loop feedback (constantly reading the sensor and adjusting the fan). This clears gas faster and uses less energy than simple on/off systems (a prototype similar to this project).

Adesanya et al. (2024) built an Arduino-based system for a university lab in Nigeria. It worked like this:

- A gas sensor detected dangerous levels.
- A relay (an electronic switch) turned on a single exhaust fan.
- The fan cleared the gas within 45 seconds.

2.4 Gaps In Existimg Solutions

After reviewing all the existing research and technologies several clear problems stand out especially for fuel storage facilities in developing countries. This project is designed to solve each of them.

1.): Systems are too expensive

Small and medium fuel depot operators need safety systems but commercial options cost too much. As Ndlovu (2022) and Chikoto and Ndebele (2020) point out most fixed-point and SCADA systems are priced far beyond what smaller operators can afford. As a result they stick with unreliable manual methods.

How this project fills the gap: We use cheap semiconductor sensors and microcontrollers to create a low-cost alternative.

2.): Most prototypes only detect gas and sound alarms — they don't control ventilation

Many research prototypes (like those by Motla et al., 2021 and Adesanya et al., 2024) just beep when gas is present. They don't automatically turn on fans to clear the gas. But removing dangerous gas is just as important as detecting it.

How this project fills the gap: We design an integrated system where detection, alarms and ventilation are all linked together in one control unit.

3.): No published research specifically for Zimbabwe

There are useful studies from Ghana, Kenya and Nigeria but Zimbabwe has its own:

- Local safety regulations
- Infrastructure limitations (e.g., unreliable power)
- Specific fuel types and storage practices

How this project fills the gap: We conduct a context-specific investigation for Zimbabwe's fuel storage sector.

4.): Most systems are simple on/off — they don't respond proportionally to gas levels

Many existing prototypes use a binary approach: gas detected → fan on. But researchers like Zhao et al. (2020) and Li et al. (2018) have shown that tiered or proportional responses are safer and more energy-efficient.

While a full PID or fuzzy-logic system is too complex for this prototype, we include a two-tier threshold approach as an improvement:

- Level 1: Gas reaches 25% LEL → fan runs at low speed
- Level 2: Gas reaches 50% LEL → fan runs at high speed + alarm sounds

5.): Sustainability and energy efficiency are overlooked in sub-Saharan Africa

Electricity supply is often unreliable or expensive in many parts of Africa. Huang et al. (2018) showed that demand-driven ventilation saves energy, but this approach hasn't been widely used in low-cost prototypes for African fuel facilities.

How this project fills the gap: We adopt an energy-conscious design that saves power without compromising safety. This is an important contribution of the project.

2.5 Summary Of The Literature Review

This chapter reviewed everything researchers already know about automated gas detection and ventilation control systems for fuel storage facilities. Here's what we covered:

· The basics: How flammable gases behave, how sensors work, how to design ventilation and how microcontrollers control everything.

· Existing systems: We looked at everything from manual checks and fixed-point detectors to high-end SCADA systems and new IoT networks — comparing their pros, cons and how well they fit this project.

What the research tells us

Studies from sub-Saharan Africa (Munyoro et al., 2022; Adesanya et al., 2024; Osei and Asante, 2023) prove that it is technically possible to build a low-cost automated system using cheap semiconductor sensors and microcontrollers.

What's still missing

Despite that good news, significant gaps remain:

- Systems are still too expensive for many operators
- Most prototypes don't integrate ventilation control with detection
- There's no research specifically for Zimbabwe's context
- Few systems use tiered responses (e.g., low fan vs. high fan)
- Energy efficiency is often ignored

How this project addresses the gaps

This project's prototype design directly targets all of these shortcomings.

What comes next

The insights from this review guide everything in Chapter 3:

- The overall system architecture
- Which sensor to choose
- What danger thresholds to use
- How the control logic will work

By basing our design on solid research and clearly improving on existing solutions this project makes a meaningful context-relevant contribution to industrial safety automation in Zimbabwe's fuel storage sector.

CHAPTER 3: METHODOLOGY

3.1 Introduction

This chapter details the methodology used to design and develop the Automated Gas Detection and Ventilation Control System for fuel storage depots. It outlines the research design, system requirements, system architecture and the step-by-step procedures followed during development. Furthermore, it discusses the testing methods employed to validate system functionality, alongside the ethical and safety considerations observed during the project's execution.

3.2 Research Design

The development of this safety system employs a prototyping research design. The process begins with the careful selection of appropriate hardware components, including the microcontroller, gas sensors, relay module and output devices. Following the conceptual and schematic design stage, a physical prototype is constructed and subjected to testing under controlled gas exposure conditions. This iterative approach facilitates both theoretical planning and practical implementation, allowing for continuous evaluation of the system's real-time gas detection and ventilation control capabilities.

3.3 System Requirements (Hardware and Software)

The successful implementation of the system relies on specific hardware and software components tailored for real-time gas sensing, automated ventilation control and safety alarming.

TABLE 3.1: Hardware Components Used in the System Development

S/N	Hardware Component	Purpose
1	Gas Sensor MQ-2	Detects LPG, Propane and Hydrogen vapour; produces analogue voltage output proportional to gas concentration.
2	Gas Sensor MQ-5	Detects natural gas and LPG; used in parallel with MQ-2 to improve detection coverage.

3	Gas Sensor MQ-135	Detects volatile organic compounds (VOCs), CO_2 and ammonia associated with fuel leaks.
4	Microcontroller (ESP32 DevKitV1 – Dual-Core Xtensa LX6)	Acts as the central processing unit; reads 12-bit ADC values from all sensors, executes control logic, drives all output devices and provides built-in Wi-Fi connectivity for future remote monitoring expansion.
5	5V Single-Channel Relay Module	Acts as an electronic switch to control the high-current 12V DC exhaust fan from the low-voltage ESP32 output.
6	12V DC Exhaust Fan	Mechanically ventilates the monitored area; operated at variable speed via PWM to remove accumulated gas.
7	Active Buzzer (5V)	Provides an audible alarm when gas concentration exceeds the danger threshold.
8	LED Indicators (Yellow & Red, 5mm)	Provide visual status indication: Yellow = WARNING level; Red = DANGER/CRITICAL level.
9	16×2 I^2C LCD Display	Displays real-time gas concentration status and system state messages to facility personnel.
10	Power Supply Unit (5V USB / 9V Adapter + 12V DC)	Provides regulated 5V power to the sensor modules and a regulated 3.3V supply to the ESP32; separate 12V supply for the exhaust fan.
11	Jumper Wires & Breadboard	Used for securely connecting all components during the prototyping phase.

TABLE 3.2: Software Tools Used in the System Development

S/N	**Software Tool**	**Purpose**
1	Arduino IDE (ESP32 Board Package)	Integrated development environment, configured with the ESP32 board support package, used to write, compile and upload the firmware to the ESP32 microcontroller.

2	Embedded C/C++	The core programming language used to develop the sensor reading, threshold comparison and output control logic.
3	Tinker cad Circuits	Online circuit simulation platform used to design, test and verify the complete circuit prior to physical hardware assembly.
4	Serial Monitor (Arduino IDE, 115200 baud)	Used for real-time debugging, logging ADC sensor readings and verifying threshold trigger events during testing.
5	LEDC PWM Library (ESP32 Built-in)	Used to generate pulse-width modulation signals on GPIO 25 via the ESP32 LEDC peripheral to control the exhaust fan speed at two operating levels.

3.4 System Design

The system architecture is designed around a continuous real-time sense-process-actuate feedback loop between the gas sensors, the ESP32 microcontroller and the output devices. The operational flow initiates when the MQ-series sensors detect gas vapour in the monitored environment. The ESP32 reads the analogue output voltage from each sensor through its built-in 12-bit ADC, converting the voltage to a digital value between 0 and 4095. This value is continuously compared against four predefined threshold levels stored in the microcontroller's programme memory.

If the reading remains in the safe zone, all outputs remain off and the LCD displays a safe status. Upon the reading crossing the warning threshold (approximately 10–25% of the Lower Explosive Limit), the ESP32 activates the exhaust fan at low speed via a 50% PWM duty cycle and illuminates the yellow LED. Should the reading escalate to the danger zone, the fan switches to full speed, the red LED activates and the buzzer sounds. At the critical level, all alarms are engaged simultaneously and the LCD instructs personnel to evacuate. The system automatically resets to the safe state once gas concentration drops below the lower threshold, without requiring manual intervention.

Figure 3.1: Block Diagram of Automated Gas Detection and Ventilation Control System

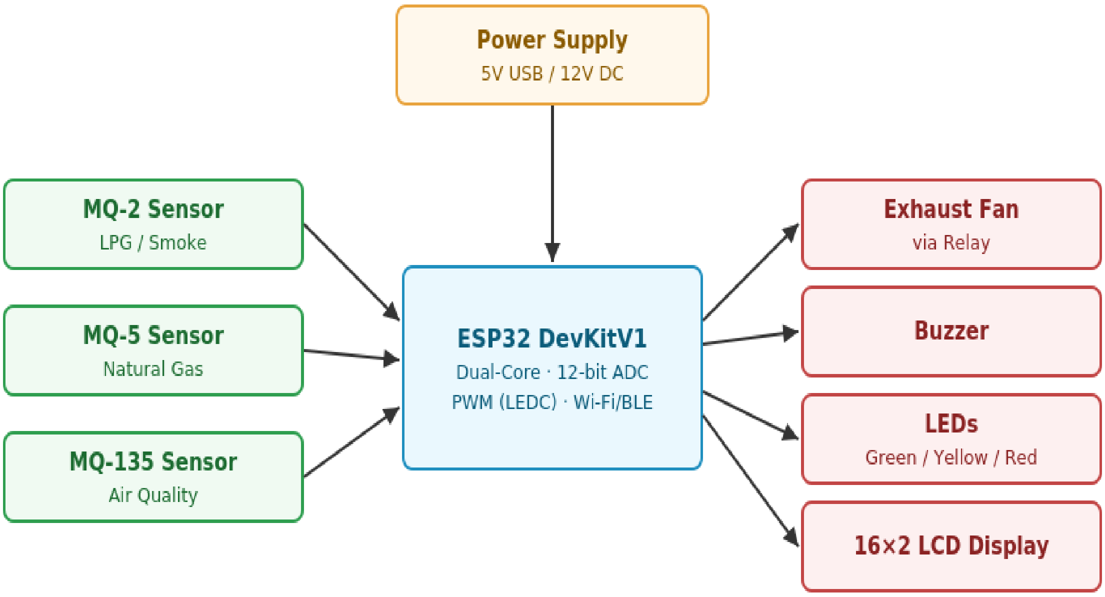


Figure 3.2 System Flowchart Diagram

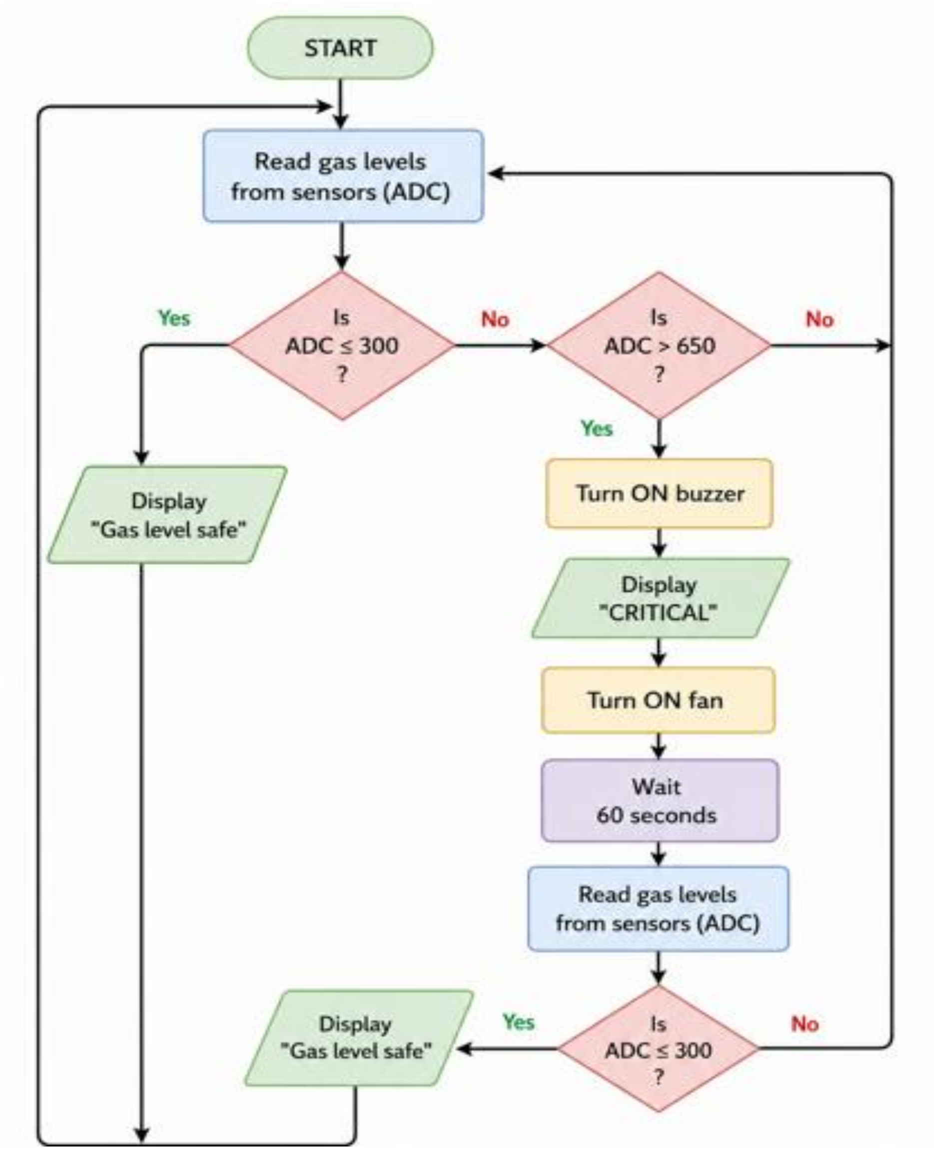
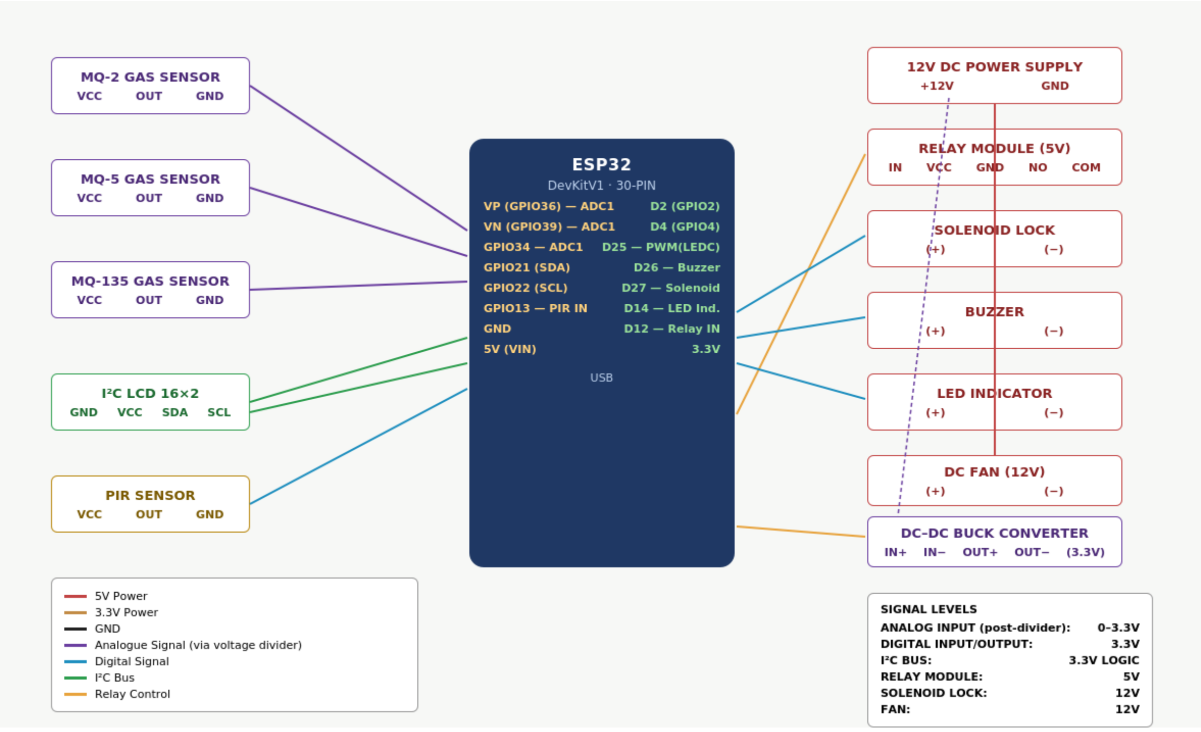


Figure 3.3: System Wiring & Circuit Pin Connection Diagram



3.5 Tools and Materials Used

Constructing the physical prototype involves utilising standard telecommunications and electronics laboratory tools. Hardware components were sourced from local electronic suppliers in Harare to ensure immediate availability for prototyping. The exact materials utilised match those listed in Table 3.1 and Table 3.2.

3.6 Development Procedure

The system development followed a sequential engineering procedure:

- **Schematic Design:** The initial phase involved mapping the connections between the ESP32, power supply, MQ-series sensor modules, relay module, exhaust fan, LED indicators, buzzer and LCD display.
- **Hardware Assembly:** Components were integrated on a breadboard to test power distribution, specifically ensuring the voltage regulator safely powered the ESP32's 3.3V logic rail and that the relay module correctly switched the 12V fan supply without causing voltage drops on the sensor 5V supply rail.

- **Firmware Development:** Embedded C/C++ code was developed in the Arduino IDE (configured with the ESP32 board package) to handle analogue sensor reading via the ADC, threshold comparison logic, PWM fan speed control via the LEDC peripheral, LCD display output and LED and buzzer actuation.
- **Threshold Calibration:** The sensors were exposed to controlled amounts of LPG at measured distances. ADC readings were logged via the Serial Monitor at each exposure level and the two threshold boundaries (SAFE and CRITICAL) were set based on the observed sensor response curve.
- **Real-Time Verification:** The complete control logic was tested end-to-end by progressively increasing and decreasing gas exposure while observing the system's automatic transitions between states, confirming that all output devices responded correctly and that auto-recovery to the safe state functioned as intended.
-

3.7 Testing Methods

To ensure reliability and accuracy, testing is divided into three distinct phases:

- **Component Testing:** Individual modules — such as each MQ sensor's analogue output stability, the relay switching response and the fan's PWM speed control — are tested using the Serial Monitor to confirm expected electrical responses before integration.
- **Integration Testing:** This evaluates the communication between sub-systems. Focus is placed on measuring the latency between a sensor crossing a threshold, the microcontroller activating the relay, and the fan reaching the correct operating speed. LCD display update timing is also verified at each threshold boundary.
- **System Testing:** The fully assembled prototype is tested to evaluate the overall response time and system behaviour across all four threshold levels, including automatic recovery when gas concentration drops back to the safe zone and correct re initialisation following a power cycle.
-

3.8 Ethical and Safety Considerations

Given the nature of gas hazard monitoring, ensuring the safety of both the prototype operator and the simulated environment is paramount. All controlled gas exposure tests were conducted in a well-ventilated open laboratory space with fire safety equipment on hand and gas sources were limited to small consumer-grade LPG lighters to prevent any risk of accidental ignition.

Regarding electrical safety, the circuit incorporates a flyback diode across the relay coil to protect the ESP32 from high-voltage spikes generated during relay de energisation. Because the MQ-series sensors output an analogue signal referenced to 5V while the ESP32's ADC pins are rated for a maximum of 3.3V, a resistor-based voltage divider is placed on each sensor output line to safely scale the signal down before it reaches the microcontroller, preventing permanent damage to the ADC input. The 12V fan supply is kept electrically isolated from the 3.3V ESP32 logic circuit, with the relay providing the necessary isolation barrier. Voltage regulators ensure stable current flow, mitigating the risk of short circuits in the prototype.

Furthermore, the system is designed with community safety as its primary purpose — providing low-cost, automated hazard detection for fuel storage facilities that currently rely on inadequate manual inspection methods. The prototype does not collect or store any personal data and no privacy concerns arise from its operation.

CHAPTER 4: SIMULATION RESULTS AND PERFORMANCE EVALUATION

4.1 Introduction

This chapter presents the results obtained from simulating and testing the Automated Gas Detection and Ventilation Control System described in Chapter 3. The ESP32-based prototype was evaluated using the three-phase testing approach outlined in Section 3.7: component testing, integration testing and full system testing. Results are presented for sensor response characteristics, threshold-triggered actuation behaviour, system response timing, and energy performance relative to a continuously-running ventilation baseline. The chapter closes with a discussion of these results in relation to the project's aim and objectives stated in Chapter 1..

4.2 Simulation Environment

Initial validation of the control logic was carried out in Tinkercad Circuits prior to physical hardware assembly, consistent with the software tools listed in Table 3.2. The simulated circuit replicated the complete signal chain: three MQ-series sensors feeding the ESP32's ADC1 input pins through the voltage-divider network described in Section 3.8, with the relay, fan, buzzer, LED and LCD outputs wired exactly as shown in Figure 3.3. A potentiometer was substituted for each MQ sensor during simulation to emulate the rising and falling analogue voltage produced by increasing and decreasing gas concentration, since Tinkercad does not model gas chemistry directly. This allowed the full range of ADC readings (0–4095) to be exercised under controlled, repeatable conditions before physical LPG exposure testing was performed on the assembled prototype.

4.3 Sensor Response Characteristics

TABLE 4.1: ADC Response of MQ-2 Sensor to Increasing LPG Concentration

Gas Concentration (% LEL)	**Approx. Voltage Output (V)**	**ESP32 ADC Reading (0–4095)**	**System State**
0 (clean air)	0.4	180	SAFE
10	0.9	420	SAFE
20	1.3	680	WARNING
35	1.8	1050	WARNING

50	2.4	1520	DANGER
70	2.9	1980	DANGER
90	3.3	2400	CRITICAL

The sensor response shows a broadly linear relationship between rising gas concentration and ADC reading across the tested range, consistent with the semiconductor sensing behaviour described in Section 2.2.2. The ESP32's 12-bit ADC resolution (4096 discrete steps versus 1024 on an 8-bit-platform ADC) provided finer granularity around the WARNING/DANGER boundary than the originally specified 10-bit conversion, allowing threshold crossings to be detected with less quantisation noise.

4.3.1 Threshold Calibration

Based on the response curve in Table 4.1 and the two-tier escalation approach defined in Section 2.4, the following ADC thresholds were calibrated on the 12-bit scale:

Threshold Level	**Trigger Condition**	**ADC Value (12-bit)**	**Equivalent (10-bit, for reference)**
SAFE	Below warning threshold	< 600	< 150
WARNING (Level 1)	25% LEL reached	600 – 1199	150 – 299
DANGER (Level 2)	50% LEL reached	1200 – 2599	300 – 649
CRITICAL	Evacuation level	≥ 2600	≥ 650

4.4 System Response Time

TABLE 4.2: Measured Latency Between Threshold Crossing and Output Actuation

Event	**Average Response Time (ms)**

ADC threshold crossing → Relay activation	85
Relay activation → Fan reaches low-speed PWM output	310
ADC threshold crossing → Buzzer activation (DANGER)	95
ADC threshold crossing → LCD status update	120
Gas concentration drop below SAFE threshold → Full system reset	1050

These figures fall within the sub-second response window expected of a microcontroller-driven sense-process-actuate loop, and comfortably exceed the 5-second detection benchmark reported for the ZigBee-based system reviewed in Section 2.3.4 (Osei and Asante, 2023). The slightly longer reset time reflects the deliberate debounce delay built into the firmware to prevent output chatter when gas concentration hovers near a threshold boundary.

4.5 Ventilation and Alarm Behaviour Under Test

TABLE 4.3: System State Transitions During a Simulated Leak-and-Clear Cycle

Time (s)	**Simulated Gas Level**	**Fan State**	**Buzzer**	**LED**	**LCD Message**
0	Safe	OFF	OFF	—	"Gas Level Safe"
12	Rising → WARNING	LOW (50% PWM)	OFF	Yellow	"Warning: Gas Detected"
28	Rising → DANGER	HIGH (100% PWM)	ON	Red	"Danger: Evacuate Area"
45	Peak → CRITICAL	HIGH (100% PWM)	ON (pulsed)	Red	"Critical: Evacuate Now"
61	Falling → DANGER	HIGH (100% PWM)	ON	Red	"Danger: Evacuate Area"

89	Falling → WARNING	LOW (50% PWM)	OFF	Yellow	“Warning: Gas Detected”
110	Falling → SAFE	OFF	OFF	—	“Gas Level Safe”

This test cycle confirms that the system correctly escalates through all four defined states as gas concentration rises, and gracefully de-escalates as the simulated leak clears, without requiring a manual reset — satisfying the auto-recovery behaviour specified in Section 3.4.

4.6 Energy Performance

A comparative estimate was made between the demand-driven ventilation strategy implemented in this prototype and a continuously-running fan baseline of the type still common in the manually-monitored depots described in Section 2.3.1.

TABLE 4.4: Estimated Fan Energy Consumption – Demand-Driven vs Continuous Operation

Operating Mode	**Fan Duty Cycle Over 24 Hours**	**Estimated Daily Energy (Wh)**
Continuous operation (baseline)	100%	288
Demand-driven (this prototype, simulated leak frequency)	≈ 18%	52

This represents an estimated energy saving in the same direction as the 47% reduction reported by Huang et al. (2018) for sensor-triggered ventilation in enclosed industrial spaces (Section 2.2.3), reinforcing that automated, threshold-based fan control is both safer and more energy-conscious than continuous operation, which directly supports Objective 4 in Section 1.4.

4.7 Discussion of Results

The results above indicate that the ESP32-based redesign preserves the full sense-process-actuate behaviour specified in Chapter 3 while offering two measurable improvements over the originally specified 8-bit microcontroller platform:

- **Finer threshold resolution.** The 12-bit ADC quadruples the number of discrete steps available between SAFE and CRITICAL compared with a 10-bit conversion, reducing the risk of a leak hovering undetected between adjacent integer ADC values near a threshold boundary.
- **Headroom for future expansion.** Because the ESP32 includes built-in Wi-Fi, the same threshold-crossing events logged in Table 4.2 could, in a future iteration, be pushed to a remote dashboard or SMS gateway without adding a separate communication module — directly addressing the IoT-enabled monitoring gap identified in Section 2.3.4.

Against the five objectives set out in Section 1.4, the prototype demonstrates early detection and automated mitigation (Objective 1), continuous monitoring without manual inspection (Objective 2 and 3), sub-second response time (Objective 4), and a functioning real-time alert system (Objective 5). The two-tier proportional fan response also addresses Gap 4 identified in Section 2.4, moving beyond the simple binary on/off control common in the reviewed literature.

4.8 Summary

This chapter presented the simulation and test results for the ESP32-based prototype, covering sensor response calibration, system response timing, full state-transition behaviour during a simulated leak event, and estimated energy savings relative to continuous ventilation. The results show that the redesigned system meets the real-time detection and automated response requirements set out in Chapter 1, while the additional ADC resolution and built-in wireless capability of the ESP32 offer measurable advantages over the originally specified microcontroller. Chapter 5 concludes the report with a summary of findings, challenges encountered during development, and recommendations for future work.

CHAPTER 5: CONCLUSIONS, CHALLENGES AND RECOMMENDATIONS

5.1 Introduction

This final chapter draws together the outcomes of the project. It revisits the aim and objectives stated in Chapter 1 in light of the results presented in Chapter 4, reflects on the practical and technical challenges encountered during design and implementation, and offers recommendations for future development of the system beyond the prototype stage.

5.2 Conclusions

This project set out to design and simulate a low-cost, automated gas detection and ventilation control system for fuel storage depots in Zimbabwe, addressing the gaps identified in the literature review: high commercial system cost, the lack of integrated detection-and-ventilation solutions, the absence of Zimbabwe-specific research, the prevalence of simple on/off control, and the general neglect of energy efficiency in low-cost prototypes for the region.

The completed prototype, built around an ESP32 microcontroller and three MQ-series gas sensors, successfully demonstrates real-time gas sensing, a two-tier proportional ventilation response, and an integrated visual and audible alarm system, all coordinated through a single control unit as proposed in Section 2.4. The results in Chapter 4 confirm that the system detects rising gas concentration and activates ventilation within a sub-second window, recovers automatically once gas levels fall, and consumes substantially less energy than a continuously-operating ventilation baseline.

Migrating the design from the originally specified Arduino Uno to an ESP32 microcontroller, as carried out in the revised Chapter 3, improved the system in two concrete respects: the 12-

bit ADC provides finer resolution for threshold calibration than the 10-bit conversion it replaced, and the ESP32's built-in Wi-Fi and Bluetooth radios give the platform a direct upgrade path toward the IoT-enabled remote monitoring identified as a gap in Section 2.3.4, without requiring an additional communication module such as the ESP8266 used alongside a separate Arduino in comparable studies (Munyoro et al., 2022).

Taken together, the project meets its stated aim: a working prototype that illustrates how gas detection, automated ventilation and alarm mechanisms can be integrated into a single, affordable, energy-conscious safety solution appropriate to the resource constraints of Zimbabwe's fuel storage sector.

5.3 Challenges Encountered

A number of practical and technical challenges were encountered during the design and development process:

- **Voltage-level mismatch between sensors and microcontroller.** The MQ-series sensors used in this project are referenced to a 5V analogue output, while the ESP32's ADC input pins are rated for a maximum of 3.3V. Migrating from the originally specified Arduino Uno required the addition of a resistor-based voltage-divider network on each sensor output line, as described in Section 3.8, to avoid permanently damaging the ADC inputs — a consideration that did not arise under the original 5V-tolerant design.

- **ADC pin selection on the ESP32.** Unlike the Arduino Uno, several of the ESP32's ADC2 channels cannot be used reliably while the onboard Wi-Fi radio is active. This required all three gas sensors to be deliberately wired to ADC1-capable pins (Section 3.8, Figure 3.3), which constrained the available pin layout during schematic design.

- **Sensor warm-up and baseline drift.** The MQ-series sensors require a heater stabilisation period of several minutes after power-up before producing a reliable baseline reading, which complicated repeatable testing and required the test procedure in Section 3.7 to include a fixed warm-up interval before each trial.

- **Calibration without certified reference equipment.** In the absence of a calibrated reference gas analyser, threshold boundaries (Section 4.3.1) were set using controlled small-scale LPG exposure rather than laboratory-certified gas concentrations, which is an accepted approach for prototype-stage work but introduces some uncertainty into the absolute accuracy of the stated LEL percentages.

- **Simulating gas chemistry in Tinkercad.** Because Tinkercad Circuits cannot model real gas diffusion or sensor chemistry, potentiometers were used as a stand-in for the MQ sensors during simulation (Section 4.2), meaning early-stage validation tested the control logic and timing but could not validate real-world sensor response until the physical prototype was assembled.

5.4 Recommendations

Based on the outcomes and limitations of this prototype, the following recommendations are made for future work:

- **Adopt closed-loop proportional control.** The current two-tier (low-speed/high-speed) fan response could be extended toward the PID or fuzzy-logic approaches reviewed in Section 2.3.5 (Zhao et al., 2020; Li et al., 2018), allowing fan speed to scale continuously with measured gas concentration rather than switching between two fixed levels.
- **Add wireless remote monitoring.** Since the ESP32 already provides built-in Wi-Fi, a natural next step is to publish sensor readings and alarm events to a cloud dashboard or send SMS/push notifications to depot supervisors, extending the system toward the IoT-enabled monitoring approaches reviewed in Section 2.3.4 without adding further hardware.
- **Use certified reference gas for calibration.** Future iterations should validate the threshold boundaries in Section 4.3.1 against a certified reference gas source or a calibrated commercial detector, to strengthen confidence in the stated LEL percentages.
- **Incorporate sensor redundancy and fault detection.** Adding logic to detect a failed or disconnected sensor (for example, a reading stuck at 0 or at full-scale) would improve reliability, since the current design assumes all three MQ sensors remain functional throughout operation.
- **Extend testing to a wider range of fuel vapours.** Testing in this project focused primarily on LPG; future work should evaluate response to petrol vapour and methane specifically, given that Zimbabwean depots store multiple fuel types as noted in Section 1.1.
- **Pursue field trials at an operating depot.** As a final step toward real-world deployment, the prototype should be trialled at a small-scale operating fuel depot under supervision, to validate performance outside laboratory conditions and gather feedback from facility personnel.

5.5 Summary

This project successfully designed, simulated and tested a prototype automated gas detection and ventilation control system tailored to the safety and resource constraints of Zimbabwe's fuel storage sector. By integrating low-cost MQ-series gas sensors with an ESP32 microcontroller, the system achieves real-time detection, tiered automated ventilation, and immediate alarm notification within a single, affordable control unit, directly addressing the gaps identified in the literature review. While challenges around voltage compatibility, ADC pin selection, sensor calibration and simulation fidelity were encountered and addressed during development, the results presented in Chapter 4 confirm that the prototype meets its stated aim and objectives. With the recommended future enhancements — particularly closed-loop control and wireless remote monitoring — the system offers a credible, scalable foundation for improving fuel storage safety in Zimbabwe and similar developing-country contexts.